

INDRAPRASTHA INSTITUTE OF TECHNOLOGY

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LAB 5(1)

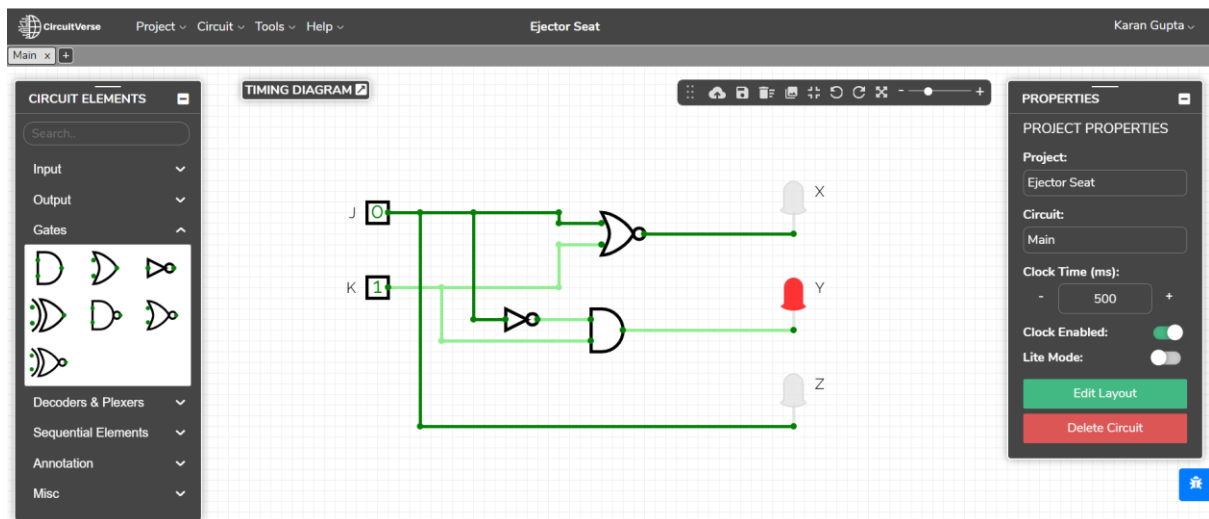
Aim: To design the logic circuit required to select the mode of ejector seats in aeroplane, modeling the orientation, air pressure and mode selector.

Components Used:

1. Input
2. NOR gate
3. NOT gates
4. AND gate
5. 3 Digital LEDs
6. Connecting wires

Link of Circuitverse: <https://circuitverse.org/simulator/edit/ejector-seat-c68472a1-9da8-480c-ae89-1ff5cd0ddea7>

Circuit Diagram:



Truth Table:

J	K	X	Y	Z
0	0	1	0	0
0	1	0	1	0
1	0	0	0	1
1	1	0	0	1

Boolean Expression: $X = J' \cdot K' = (J + K)'$
 $Y = J' \cdot K$
 $Z = J \cdot K' + J \cdot K = J \cdot (K' + K) = J$

Observation:

Mode X is on when aircraft is upright($J=0$) and air pressure is high($K=0$)

Mode Y is on when aircraft is upright($J=0$) and air pressure is low($K=1$)

Mode Z is on when aircraft is upside down($J=1$) regardless of the air pressure

Result: The ejector seats with 3 modes were designed using gates.

LAB 5(2)

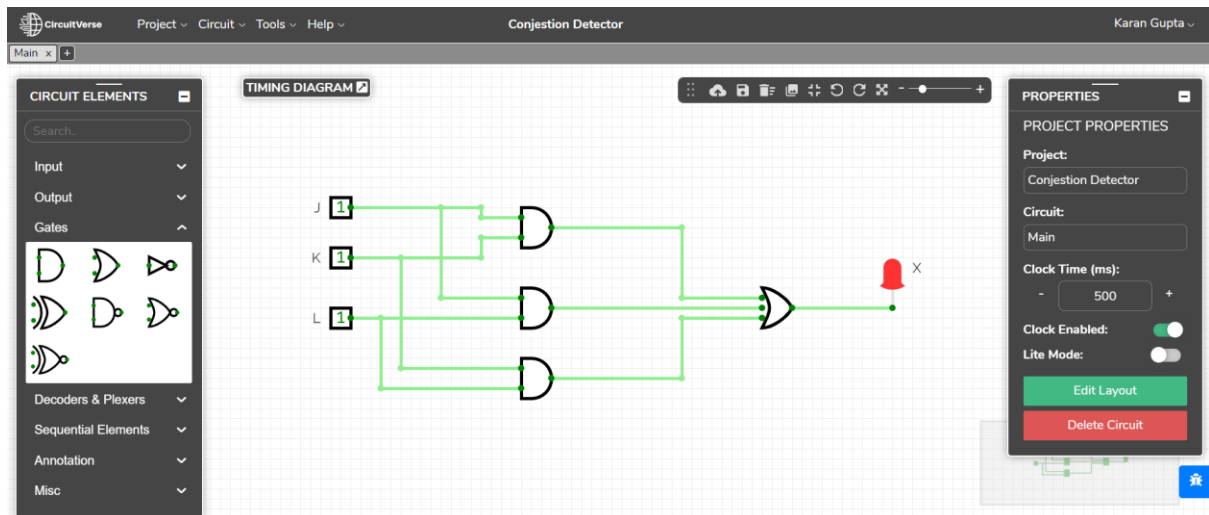
Aim: To design the logic circuit for a congestion detection system, modeling the sensors and logging device

Components Used:

1. Inputs
2. 3 AND gates
3. Connecting wires
4. LED

Link of Circuitverse: <https://circuitverse.org/simulator/edit/lab-5-b-f1f2655f-8a17-447b-bf4e-2dff5af3301c>

Circuit Diagram:



Truth Table:

J	K	L	X
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

Boolean Expression: $X = J.K + K.L + J.L + J.K.L = J.K + K.L + J.L(1 + K) = J.K + K.L + J.L$

Observation:

The logging device($X=1$) is activated when at least 2 of the sensors(J,K,L) are High.

Result: The congestion detection system was designed using gates.